# How Brains Learn

Name _____

Work with a partner. Compare answers when you finish each problem.

**Word bank:** conditioning food sound stronger weight wire

*You can use a word more than once.*

**1. Your other hand.** Write your name with your other hand. Your partner times you.

| | Try 1 | Try 2 | Try 3 |
|---|---|---|---|
| **Time (seconds)** | | | |

**2. Stronger connections.**

When you practice, the synapses between your neurons get _____.

“Neurons that fire together, _____ together.”

**3. Pavlov’s dogs.**

Pavlov’s team played a _____ right before they gave the dogs _____.

Soon the dogs drooled when they heard the sound, even with no _____.

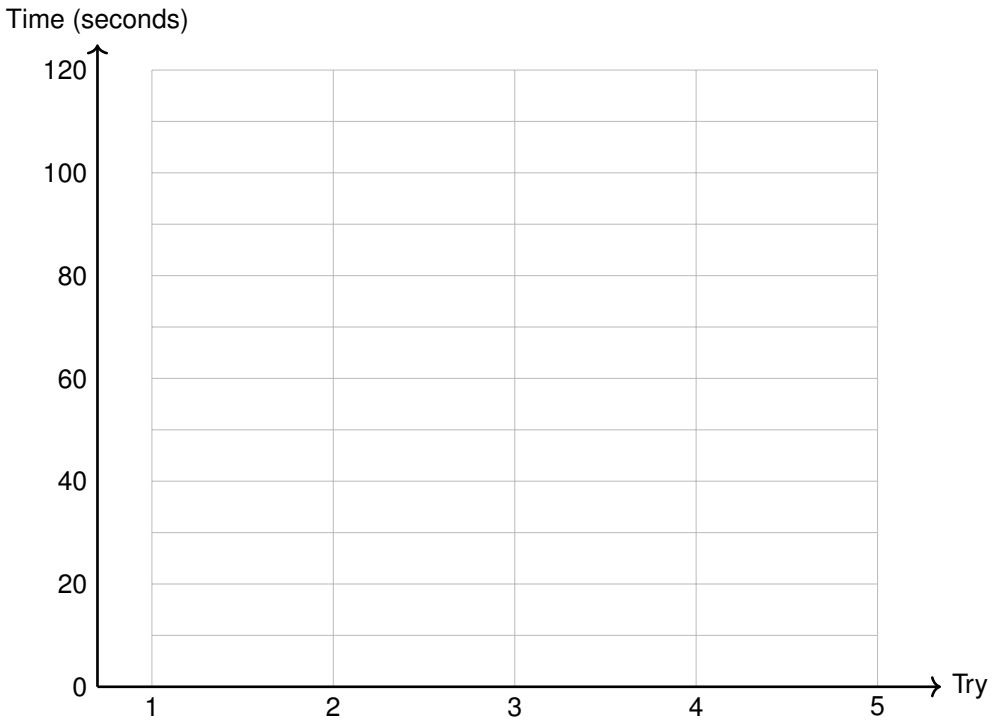
Learning to connect two things that happen together is called _____.

In AI, the strength of a connection is a number called a _____.

**4. Mirror drawing.** Use the same pen for all 5 tries.

| | Try 1 | Try 2 | Try 3 | Try 4 | Try 5 |
|---|---|---|---|---|---|
| **Time (seconds)** | | | | | |
| **Mistakes** | | | | | |

**5. My learning curve.** Put a dot for each try. Connect the dots.




What happened to your time? _____

**6. Talk about it.** Your brain changes synapses. AI changes weights. Is it the same kind of learning?

Brain learning and AI learning are **the same** / **different** because _____

_____

A sound that makes me reach for my phone: _____ That is like Pavlov's dogs.

_____

**Go Further (Day 2).** A new mirror: left and right flipped.

Same pen as yesterday: _____ seconds. New mirror: _____ _____ _____ seconds.

Was it hard again? _____ Why? _____